



YOUR TECHNOLOGY & INNOVATION PARTNER

RAG

From Foundation to Application

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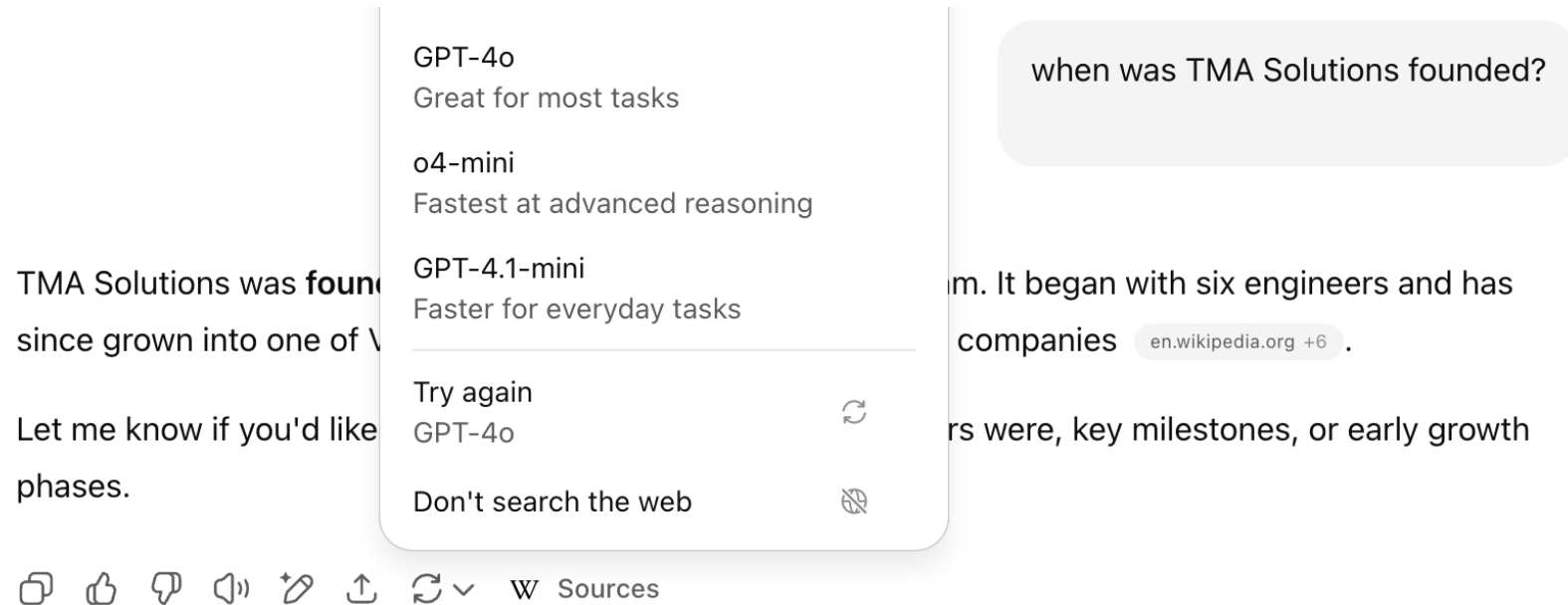
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1. Introduction to RAG

What is RAG and how it is used?

Large Language Models

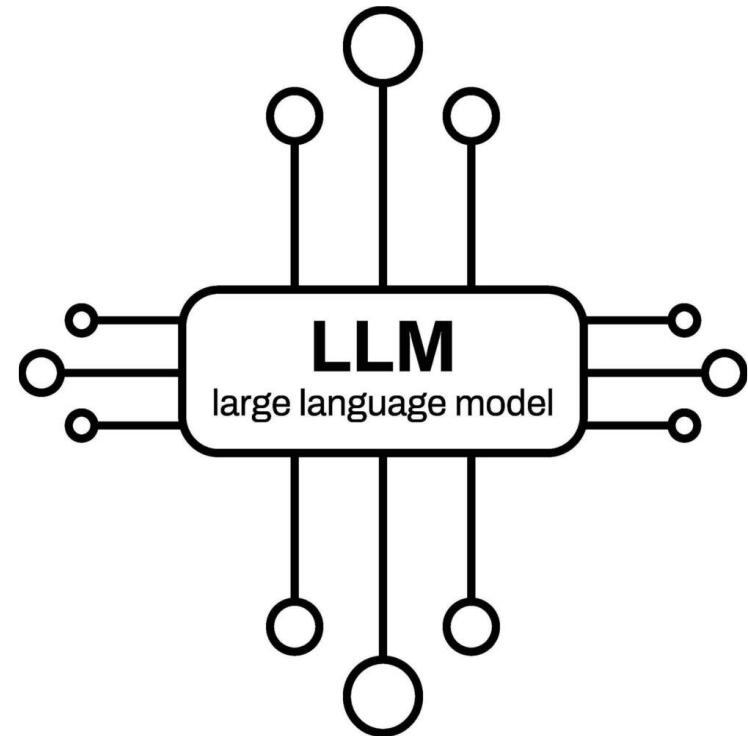
Large Language Models (LLMs) are advanced AI models trained to understand and generate human-like language.



When we interact with ChatGPT or Gemini, we're actually communicating with the language model behind them.

Limitation with LLMs (1/2)

If I have a dataset and want the LLM to answer only based on it, simply feeding the question into the LLM won't work.



Limitation with LLMs (2/2)

An example of how an LLM might give a convincing but incorrect answer.

This issue is known as **AI hallucination**.

Give answer only. What is limited size when we send and receive TMA email? ^

Select one:

- a. 40 MB
- b. 50 MB
- c. 30 MB
- d. 20 MB
- e. 10 MB



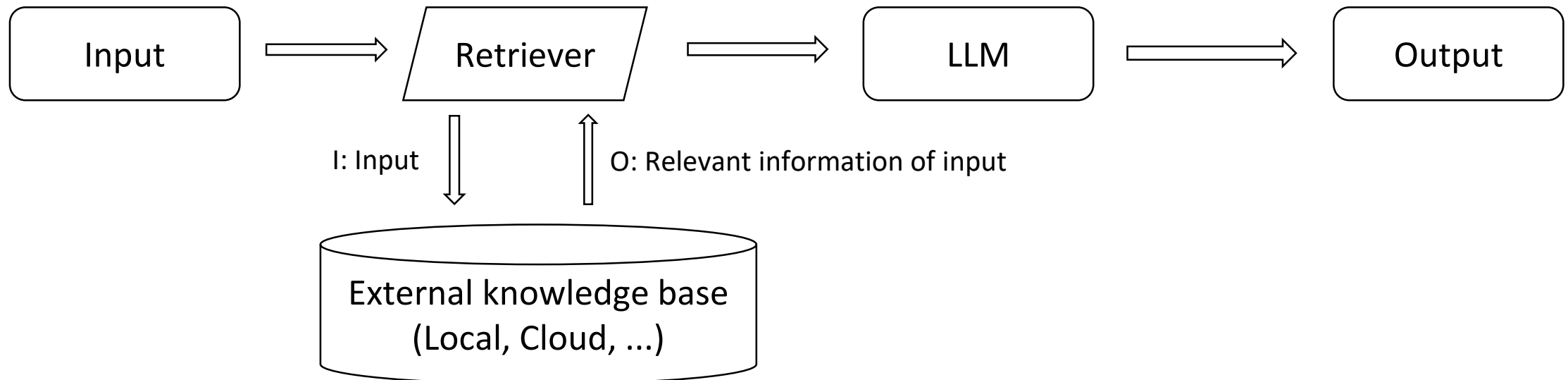
The correct answer is **b. 50 MB**. The limited size for sending and receiving TMA emails is 50 MB. This is a common limit for email attachments and overall message size in many systems.

Using retrieval to overcome LLMs' limitation

Instead of this, which might end up with the hallucination of LLMs:



Retriever component is added to get the relevant information of input from an external knowledge base:



Using retrieval to overcome LLMs' limitation

That is basically the idea of Retrieval-Augmented Generation (RAG)!

When was TMA Solutions founded?

TMA Solutions was **founded in 1997** in Ho Chi Minh City, Vietnam en.wikipedia.org +6. The company began operations with just six software engineers and has since grown into one of Vietnam's largest software outsourcing firms en.wikipedia.org +2.

So to directly answer



W Source



1/3

W en.wikipedia.org

TMA Solutions

April 22, 2025 — TMA Solutions is a privately owned software outsourcing company in Vietnam, with ...

In this example, ChatGPT used the information retrieved from Wikipedia to answer the question from user.

RAG (1/2)

RAG (Retrieval-Augmented Generation) was first introduced by Facebook AI Research (FAIR) in collaboration with UCL and NYU in 2020.

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

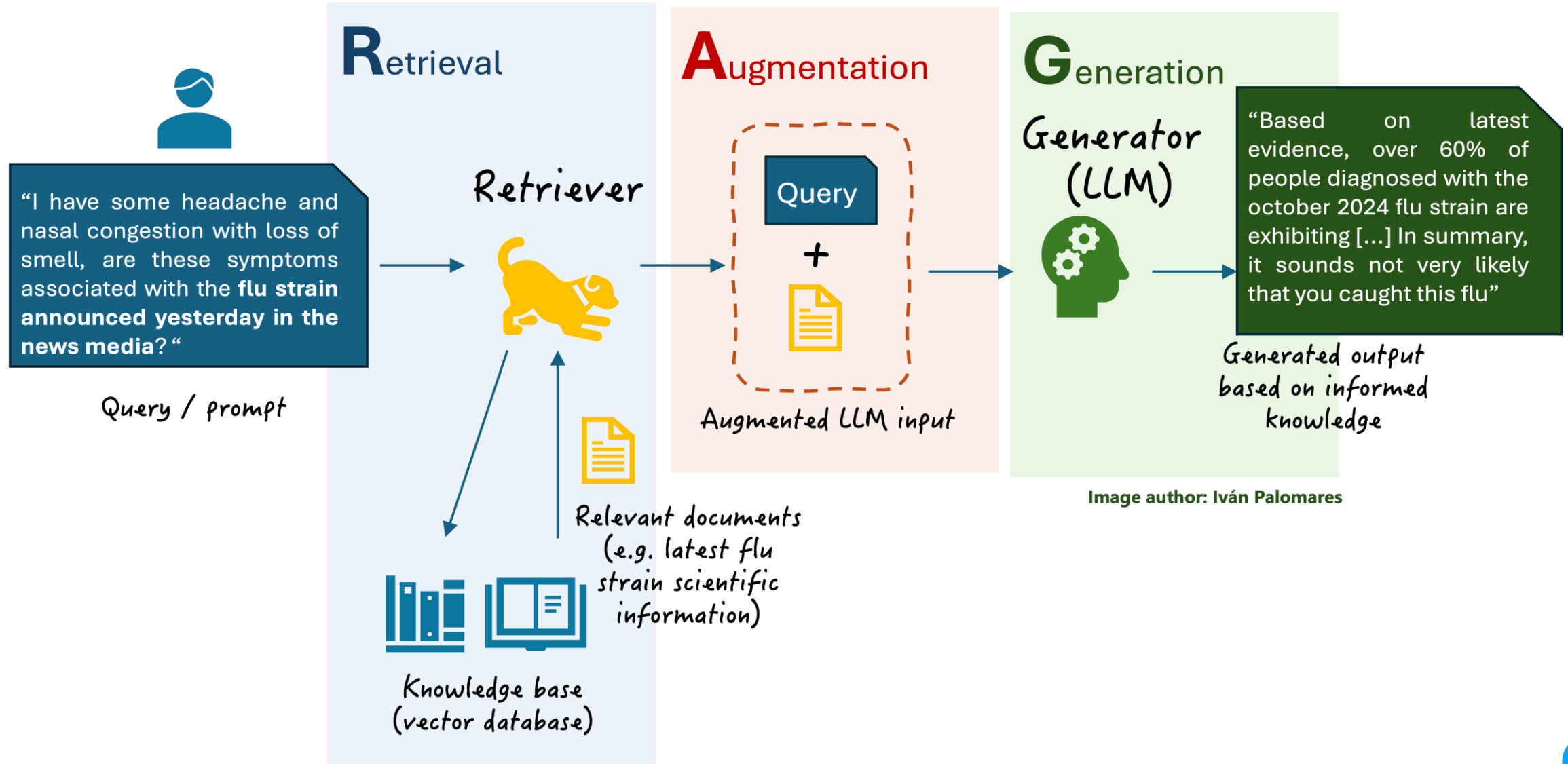
Patrick Lewis^{†‡}, Ethan Perez^{*},

Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

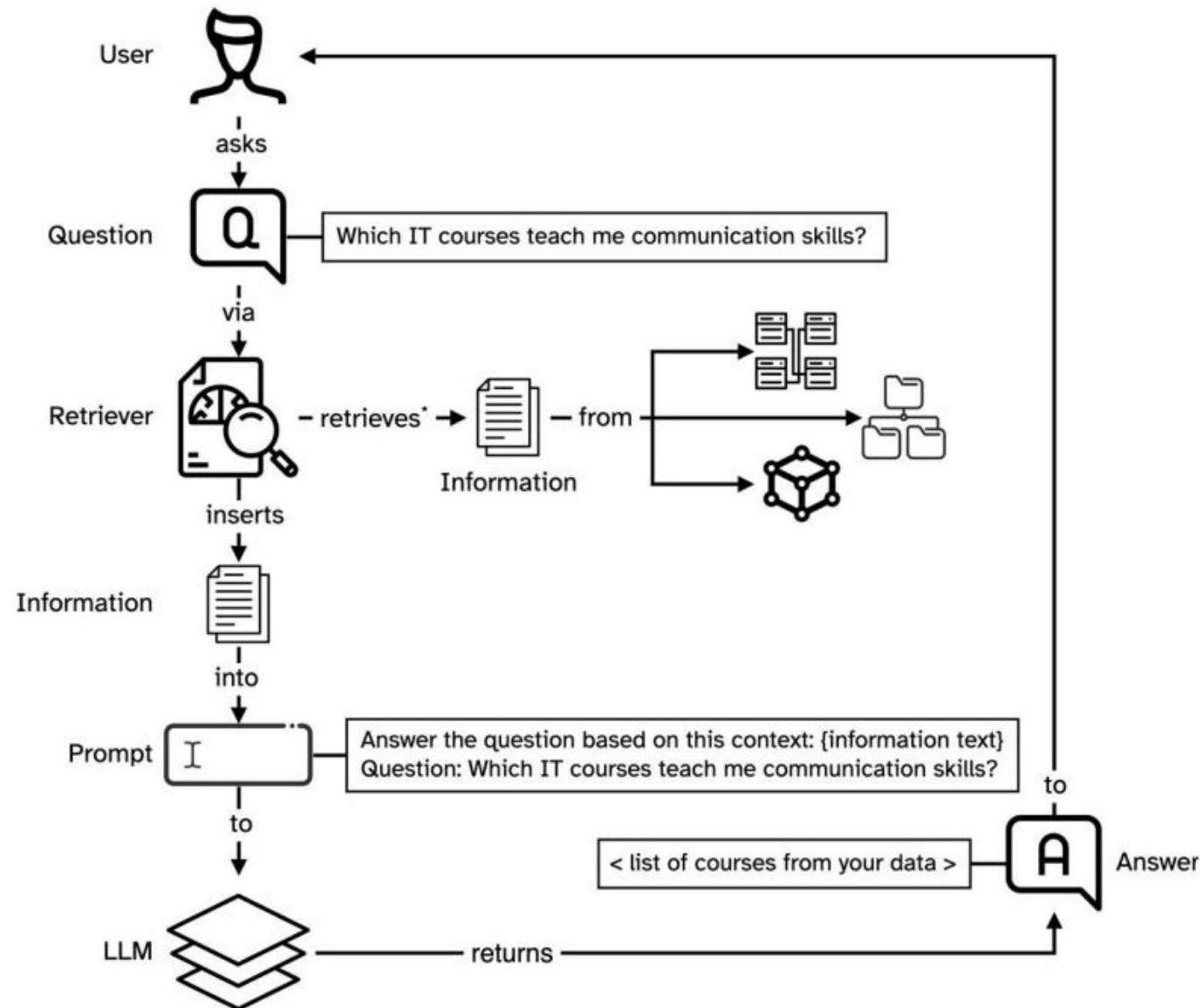
Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

[†]Facebook AI Research; [‡]University College London; ^{*}New York University;
plewis@fb.com

RAG (2/2)



RAG: An Example



AI Agent with RAG: An Example

Chat Dialog



Hello! I'm T-VA, your AI assistant from TMA Solutions. How can I help you today?



hi there



Please provide me with the two most suitable candidates for the following job description:
Introduction We are seeking an experienced Senior QA Engineer with the ability to design and implement comprehensive testing strategies for our software products. The successful candidate will play a key role in ensuring product quality, enhancing user experience, and optimizing the development process...



Here are two suitable candidates for the Senior QA Engineer position:

Candidate 1: Hung Nguyen ⓘ

Full Name: Hung Nguyen

Relevant skills and experience:

6 years of experience in Automation and Manual Test, including a technical leader role in building automation frameworks. ⓘ

Strong experience with automated testing tools like Selenium, Robot

Type a message



TMA Virtual Assistant can make mistakes. Check important info.

Case 1



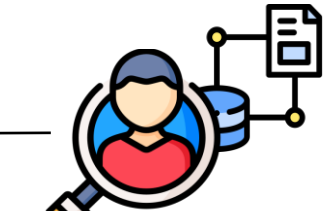
direct answer

Resume Agent doesn't need to use the retrieval tool!

Case 2



Relevant candidates



Use retrieval tool



use RAG first, before give an answer

Resume Agent uses the retrieval tool to find candidates!

AI Agents with RAG: Use Cases

There are a lot of usage of using RAG in real life!

- Internal Helper Assistant
- Customer Support Assistant
- HR Assistant

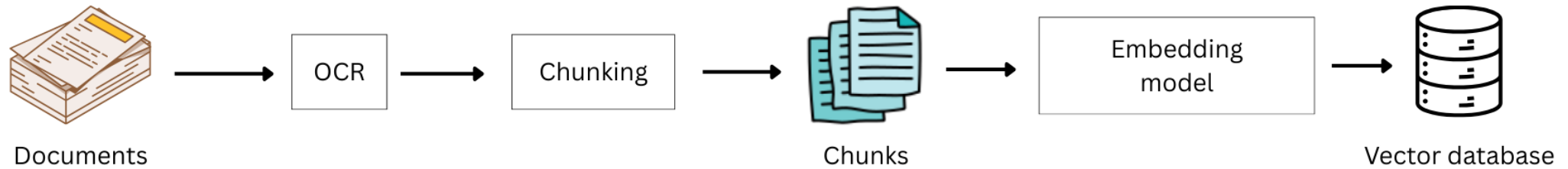
And more!

2. Anatomy of a RAG System

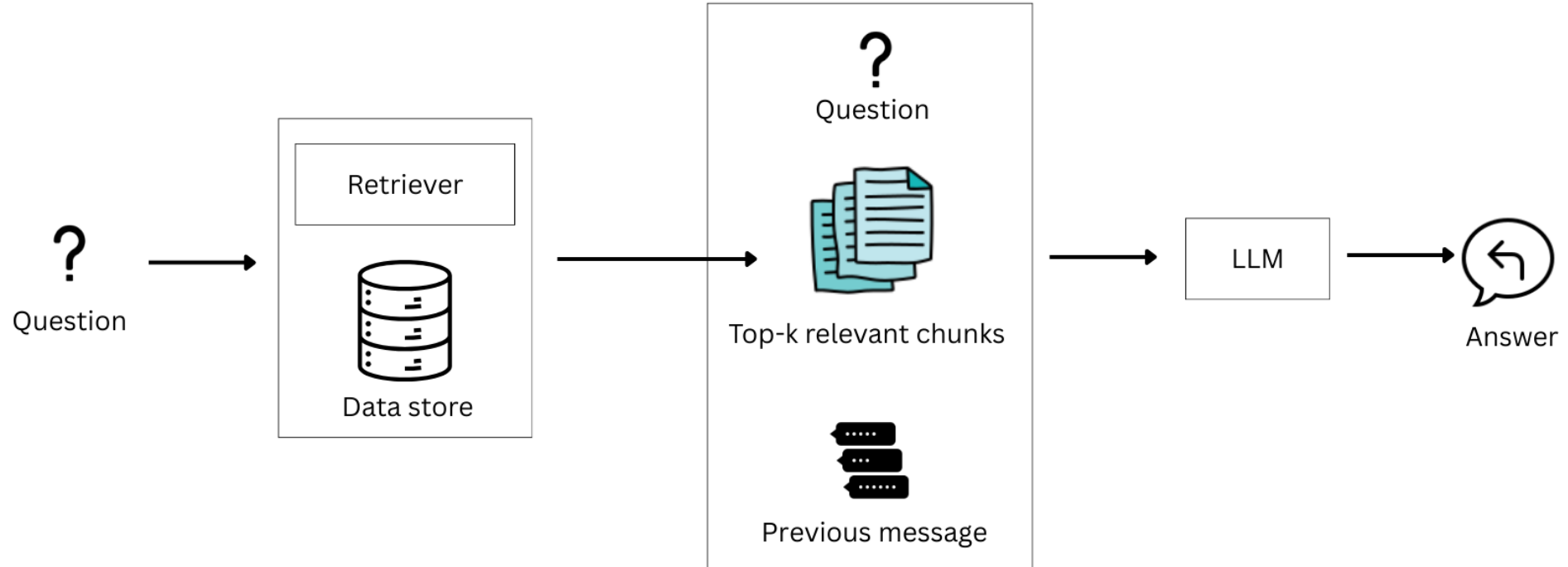
Let's see what exactly is inside a RAG system!

RAG Pipeline - Overview

Indexing



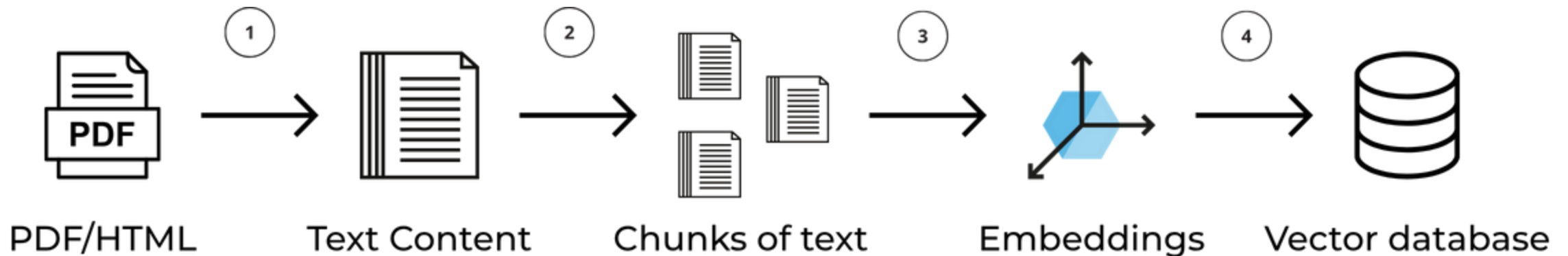
Generating



Indexing Stage - Overview

Process:

- (1) Extract the text from documents by using OCR tool.
- (2) Chunking - the process of splitting large documents into smaller, manageable text units (called chunks).
- (3) Convert chunk into a form called vector by using embedding models.
- (4) Insert the metadata of each chunk into vector database.



Chunking

Chunking method should be selected based on the type of data.

Why is Chunking important?

- Retrieval works better when docs are preprocessed into searchable units.
- LLMs have a limited context window → cannot process whole documents.

there, fellow tech enthusiasts! I've been diving deep into the world of Retrieval-Augmented Generation (RAG) lately, and I wanted to share a handy, human-written guide on the basics of RAGs

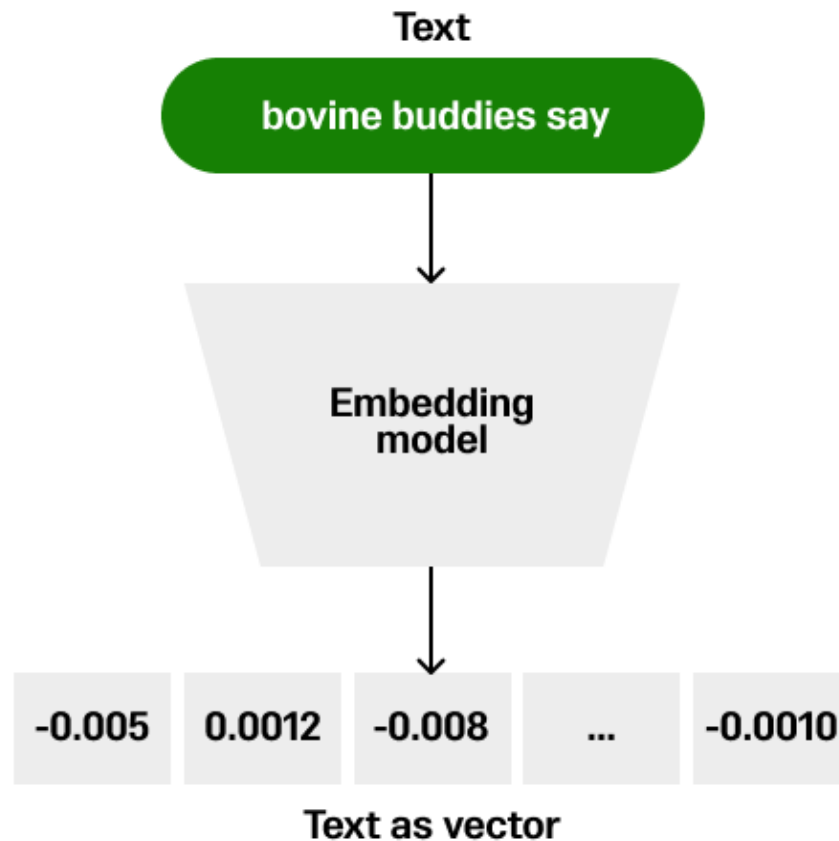
🔍 Công cụ



Tin nhắn quá dài

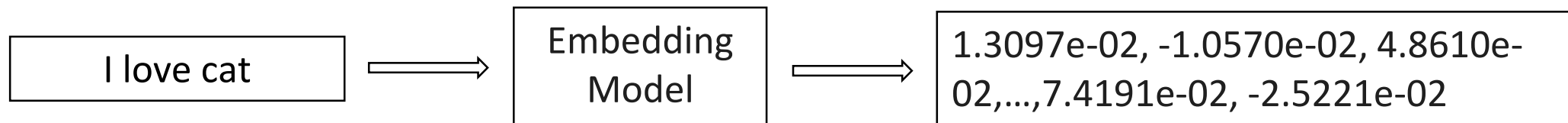
Sentence Embedding

Sentence embedding is the process of transforming a sentence into a vector of numbers that represents its meaning in a high-dimensional space.



Embedding model

An **embedding model** is a machine learning model that converts text (like a sentence or paragraph) into a fixed-length vector of numbers.



These vectors capture the semantic meaning of the text so that similar meanings have similar vectors, even if they use different words.

Vector Database

A **vector database** is a specialized data store designed to index, store, and search vectors efficiently.

In RAG, after documents are chunked and embedded, each chunk is stored in a vector database:

- Vector
- Original text
- Metadata

(e.g., title,
section, tags, ...)

```
{
  "id": "1",
  "vector": [0.142, 0.829, 0.001, 0.562, ... , 0.437, 0.923],
  "payload": {
    "filename": "TMA-Technology-Group-Booklet-VN.pdf",
    "title": "TMA Technology Group Booklet",
    "section": "Công viên sáng tạo TMA Bình Định",
    "date": "2025-08-14",
    "chunk": "Ready facility for additional thousands of engineers"
  }
}
```

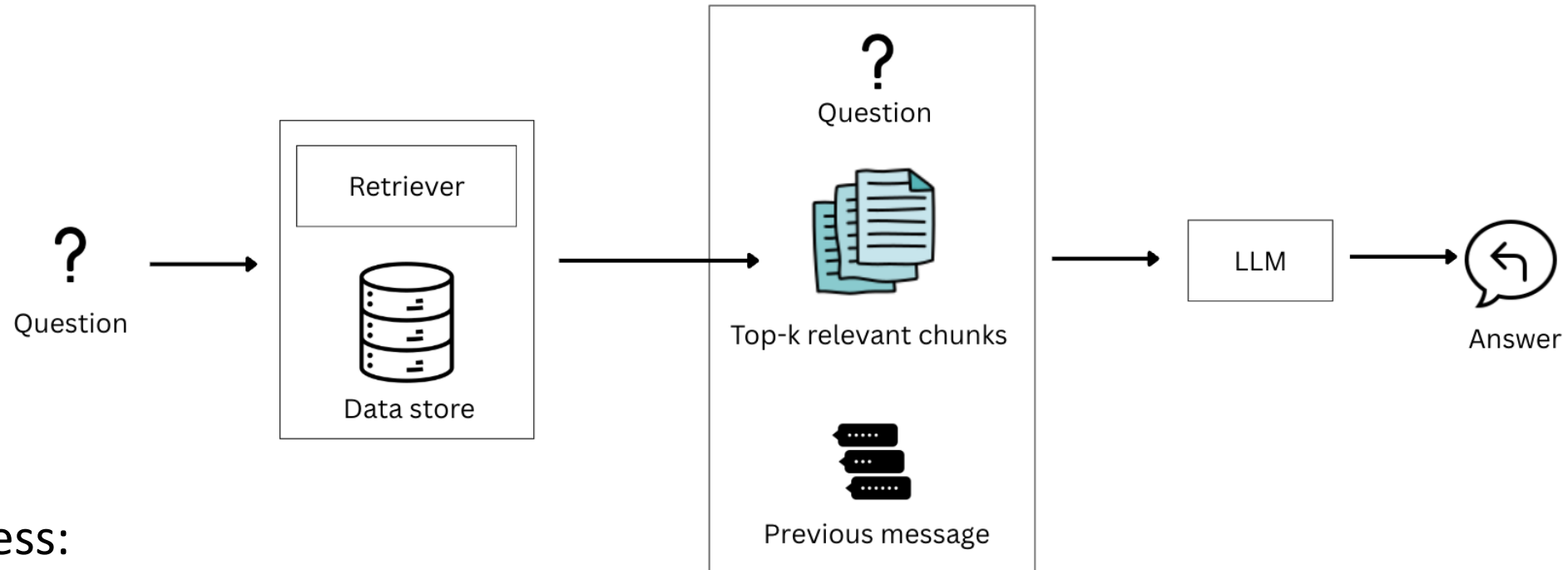
Vector Database Technologies



Chroma



Generating Stage - Overview

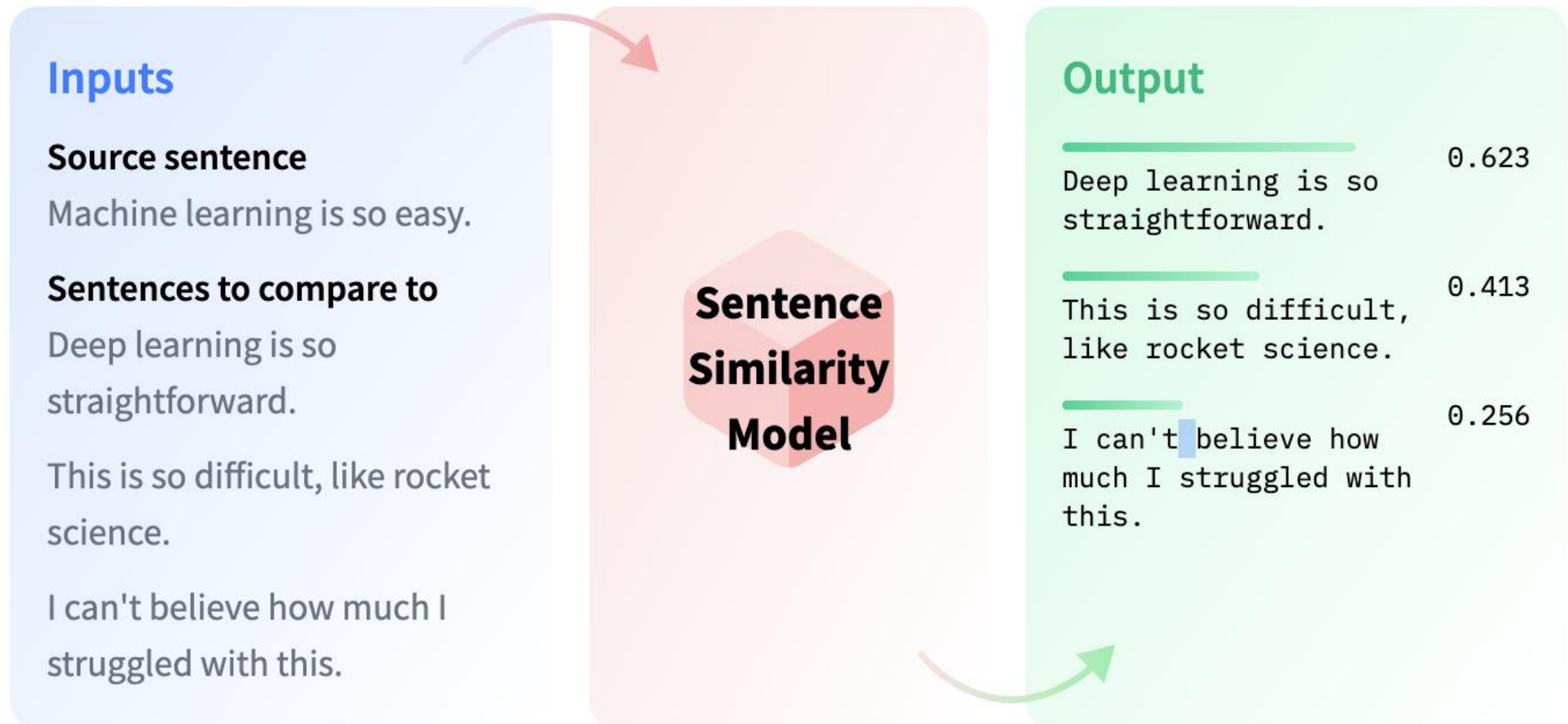


Process:

- Input are used for retrieval top relevant chunks from vector database.
- Query Q, top-k relevant chunks get from retriever and previous message (for chatbot application) are passed to LLM.
- LLM (GPT, Gemini, ...) uses them to generate an answer.

Retrieval

Problem: Suppose that we have a query Q and a chunk C, how could we know if C is relevant to Q?



Cosine Similarity (1/2)

We compare the **similarity** between the query and all stored chunks, using their vector representations.

By using embedding model, we get:

- Query Q $\implies$ Vector v_Q
- Chunk C $\implies$ Vector v_C

Source Sentence

That is a happy person

Sentences to compare to

That is a happy dog

That is a very happy person

Today is a sunny day

Add Sentence

Compute

That is a happy dog

0.695

That is a very happy person

0.943

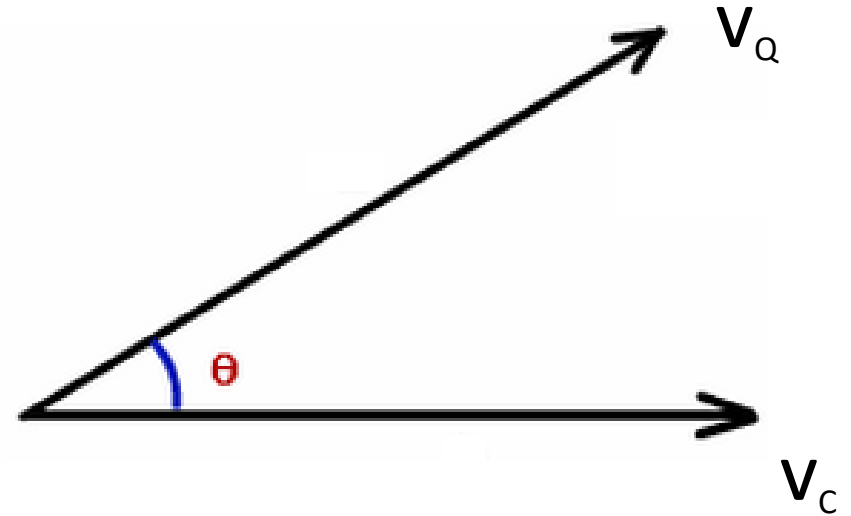
Today is a sunny day

0.257

Cosine Similarity (2/2)

Cosine similarity measures how similar two vectors are based on the **angle** between them.

$$\text{cosine_similarity}(v_Q, v_C) = \frac{v_Q \cdot v_C}{\|v_Q\| \cdot \|v_C\|}$$



Property:

- A cosine similarity closer to 1 indicates higher relevance between C and Q
- A value near 0 indicates little to no correlation.
- A value close to -1 indicates strong inverse correlation.

Let's try using an embedding model: <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

System Prompt

System Prompt is a special instruction prepended before the user's input to shape the behavior of the LLM.

Each time you write a message to any LLM (ChatGPT, Gemini, ...), they all run a system prompt behind.

Example ([Claude Opus 4](#))

```
The assistant is Claude, created by Anthropic.  
The current date is {{currentDateTime}}.  
Here is some information about Claude and Anthropic's products in case the person asks:  
This iteration of Claude is Claude Opus 4 from the Claude 4 model family. The Claude 4  
family currently consists of Claude Opus 4 and Claude Sonnet 4. Claude Opus 4 is the most  
powerful model for complex challenges.  
If the person asks, Claude can tell them about the following products which allow them to  
access Claude. Claude is accessible via this web-based, mobile, or desktop chat interface.  
Claude is accessible via an API. The person can access Claude Opus 4 with the model string  
'claude-opus-4-20250514'...
```

Prompt Construction

In a RAG pipeline, we **build a single input prompt** that contains:

- **System Prompt** – sets the assistant's behavior
- **Retrieved Chunks** – factual context to ground the answer
- **User Question** – the original query from the user

Example Prompt Template for HR Agent:

```
You are a helpful HR assistant. Use only the information provided in the context below to answer the question. If the context does not contain enough relevant information, say "I don't know" and do not speculate.
```

```
Context:
```

```
{ 1. Chunk A: "...text from employee handbook..."  
  2. Chunk B: "...text from company policy..." }
```

```
Question: {What is the company's policy on remote work?}
```

3. Create a simple RAG applications with T-VA

Let's see how RAG are being used in real life!

4. Open Q&A

Thank you for being a part of this presentation!